Huzaifa Shahbaz

Passionate Computer Engineer & Content Creator

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Summary

Huzaifa Shahbaz is a passionate Computer Engineer, lifelong learner, and content creator dedicated to advancing knowledge in computing and technology. He bridges academic theory and real-world application, empowering students, professionals, and tech enthusiasts. As a YouTuber, he produces clear, engaging computer engineering lectures, programming tutorials, and technology guides for BS, MS, and PhD students. On LinkedIn and CVs, he highlights expertise in software development, system design, and emerging technologies while aligning with PEC's professional standards. Outside tech, Huzaifa actively enjoys sports — including cricket, football, gym, badminton, volleyball, table tennis, swimming, and field tennis — and enriches his perspective through reading books, watching documentaries, and appreciating English films. His blend of analytical rigor, creative exploration, and balanced pursuits inspires academic growth and lifelong excellence.

Skills

Click Up, Online Documentation, Research Tools, Linux, Ubuntu OS, MS Visio, Figma, Smart Systems and Emerging Technologies, Applied Engineering and Innovation, Team Collaboration in Multidisciplinary Engineering Projects, Technology Evaluation and Prototype Development, System Architecture Design, MySQL Database Design and Management, Xilinx Vivado (Digital Design and FPGA Workflow), Android Studio, Visual Studio Code, Verilog HDL (Digital Logic and Processor Design), C/C++ (Embedded and Sy

Work Experience

TRAINEE R&D Sep 2025 - Nov 2025

MACHADEV ENGINEERING PVT LTD | ISLAMABAD/PAKISTAN

I work as research and development and my project was smart water monitoring anf purification device.In this I made thw water filter plant at industrial level with ESP32-S3 microcontorller and five sensors.

INTERN Jul 2025 - Aug 2025

MACHADEV ENGINEERING PVT LTD | ISLAMABAD/PAKISTAN

I DONE MY INTERNSHIP AT MACAHEV ENGINEERING PVT LTS.WHERE I HAD LEARNED EMEBEDED SYSTEMS,PCB DESIGN,SIMULATION AND TESTING AND ASLO SOLIDWORKS OF MECHNICAL ENGINEERING

INTERN Jul 2024 - Sep 2024

RISETECH | Rawalpindi Pakistan

IN THIS INTERNSHIP I HAD LEARNT MANY NEW THINGS SUCH AS ESP32-S3,STM32CUBE,8051 MICRONTROLLER AND LoRa AND LoRaWAN CONTROLLER AND ALSO ATTENDED PCB WORKSHOPS.

Education

Bachelors of Science in Computer Engineering Sep 2021 - Aug 2025

HITEC University, Rawalpindi, Pakistan

FSc(ICS)/HSSC in Science General Nov 2019 - Sep 2021

Army Public School and College EME College Campus, Rawalpindi, Pakistan

Matric SSC in Science General Apr 2017 - Nov 2019

Army Public School and College EME College Campus, Rawalpindi, Pakistan

Certifications and Training

- Construction OHS Code 2025 Need Framework & Implementation for Effective Compliance BY PAKSITAN ENGINEERIGN COUNCIL
- Data Warehousing for Smart Infrastructure, Energy & Utilities BY PAKSITAN ENGINEERIGN COUNCIL
- Project Planning & Time Management BY PAKSITAN ENGINEERIGN COUNCIL
- Reliability Engineering of Complex Systemsrends BY PAKSITAN ENGINEERIGN COUNCIL
- Smart Engineering with IoT and Cloud Integration Engineer-Entrepreneur BY PAKSITAN ENGINEERIGN COUNCIL
- SEMINAR OF ROBOTICS AND AI IN HEALTHCARE BY HITEC UNIVERSITY
- CODE WAR BY HITEC UNIVERSITY
- SPEED CODING IN ROBO FOESTA BY HITEC UNIVERSITY
- CERTIFICATE OF INTERNSHIP OF MACHADEV ENGINEERING PVT LTD BY AIR UNIVERSITY
- Huzaifa Shahbaz_Speed ProgrammingBY NUST EME COLLEGE
- CERTIFICATE OF RISETECH INTERNSHIP OF HUZAIFA SHAHBAZBY NUST EME COLLEGE

Hobbies

sports, cricket, football, gym, badminton, volleyball, table tennis, swimming, field tennis, reading books, watching documentaries, appreciating English films

Project Experience

Network Coding for Wireless Applications: A Review Project Proposal OF WIRELESS MOBILE NETWORS

Network Coding for Wireless Applications: A Review

- Conducted comprehensive literature review analyzing network coding techniques and their implementation in wireless mobile network architectures to improve throughput and reduce latency
- Evaluated performance metrics of various network coding protocols including random linear network coding (RLNC) and practical network coding (PNC) for enhancing data transmission efficiency
- Synthesized research findings on real-world applications in 5G/6G networks, IoT systems, and vehicular ad-hoc networks (VANETs) with recommendations for future deployment strategies

AI for Medical Diagnosis (Image Classification)

AI for Medical Diagnosis (Image Classification)

- Developed deep learning model using convolutional neural networks (CNN) for automated medical image classification to detect diseases from X-rays, CT scans, or MRI images with high accuracy
- Implemented transfer learning techniques with pre-trained architectures (ResNet/VGG/EfficientNet) and applied data augmentation strategies to optimize model performance on limited medical datasets
- Achieved [X%] accuracy in disease classification, demonstrating potential to assist healthcare professionals in early diagnosis and reducing diagnostic time by improving clinical decision-making workflows

G06-INDUSTRAIL-GRADE OMR SCANNER

G06-Industrial-Grade OMR Scanner – Engineering Economics Analysis

- Conducted comprehensive cost-benefit analysis and economic feasibility study for developing industrial-grade Optical Mark Recognition (OMR) scanner, evaluating capital investment, operational costs, and projected ROI over 5-year lifecycle
- Performed comparative market analysis of existing OMR solutions, identifying cost optimization opportunities through value engineering and component selection to achieve 40% cost reduction while maintaining commercial-grade accuracy
- Developed financial model demonstrating break-even analysis, NPV, and IRR calculations, presenting business case for manufacturing scalable OMR system targeting educational institutions and survey processing industries

RISETECH COMPUTER ENGINEERING DEPARTMENT NUST EME COLLEGE(ORANIZATION

RISETECH - Computer Engineering Department, NUST EME College

- Led health, safety, and environmental (HSE) initiatives for computer engineering labs and workshops, conducting risk assessments, safety audits, and implementing protocols to ensure compliance with university safety standards
- Coordinated technical events, workshops, and seminars for 200+ students, managing logistics, vendor relations, and budget allocation while enforcing HSE guidelines during hands-on training sessions
- Developed and delivered safety training programs covering electrical hazards, equipment handling, and emergency procedures, reducing lab incidents by implementing preventive measures and safety checklists

Lung Segmentation from Chest X-rays – Digital Image Processing

Lung Segmentation from Chest X-rays – Digital Image Processing

- Developed automated lung segmentation algorithm using image processing techniques including thresholding, morphological operations, edge detection (Canny/Sobel), and region growing to extract lung regions from chest X-ray images
- Implemented preprocessing pipeline with noise reduction filters (Gaussian/median), histogram equalization, and contrast enhancement to improve image quality and segmentation accuracy for varied radiographic conditions
- Achieved [X%] segmentation accuracy using MATLAB/Python with evaluation metrics including Dice coefficient, Jaccard index, and pixel-wise comparison, enabling precise lung boundary delineation for clinical analysis

IoT Based Air Quality Index Meter System (FYP): ESP32, MQ sensors, Firebase, real time monitoring.

IoT-Based Air Quality Index (AQI) Meter System – Final Year Project

- Designed and developed real-time air quality monitoring system using ESP32 microcontroller integrated with MQ-135, MQ-5, and PM2.5 AND PM10 sensors to measure CO2, CO, temperature, humidity, and pollutant levels with data logging to Firebase cloud database
- Implemented IoT architecture with WiFi connectivity enabling remote monitoring through web dashboard and mobile application, providing instant AQI calculations and pollution alerts based on WHO/EPA standards
- Deployed functional prototype achieving 95%+ sensor accuracy, demonstrating practical solution for environmental monitoring in smart cities, homes, and industrial settings with scalable cloud infrastructure

Serial port communication with Blink LEDs EMBEDDED SYSTEM

Serial Port Communication with Blink LEDs – Embedded Systems

- Implemented UART serial communication protocol on microcontroller (Arduino/STM32/8051) to establish bidirectional data transfer between embedded system and host computer, controlling LED states through serial commands
- Developed firmware in C/C++ to parse received serial data, execute command-based GPIO operations for LED blinking patterns, and transmit status messages back to terminal with configurable baud rates (9600-115200 bps)
- Demonstrated understanding of asynchronous communication, interrupt handling, and real-time embedded system response through successful implementation of multi-LED control with variable frequency and patterns

Automatic Vehicle Speed Controller: Arduino & Proteus simulation

Automatic Vehicle Speed Controller – Arduino & Proteus Simulation

- Designed and simulated intelligent speed control system using Arduino microcontroller with IR/ultrasonic sensors for automatic vehicle speed regulation based on predefined speed limits and obstacle detection in hazardous zones
- Developed embedded control algorithm implementing PID control and PWM signal generation to adjust DC motor speed, achieving smooth acceleration/deceleration with 90%+ accuracy in maintaining target speeds
- Validated system functionality through Proteus circuit simulation including sensor interfacing, LCD display for real-time speed monitoring, and buzzer alerts for over-speed warnings before hardware implementation

IOT-Based AQI Meter System for Real-Time Data Analysis and Decision Making OF Management and Entrepreneurship

IoT-Based AQI Meter System for Real-Time Data Analysis – Management & Entrepreneurship

- Led cross-functional team in developing IoT air quality monitoring product from concept to prototype, conducting market research identifying $2.5B+ global air quality sensor market with 8% CAGR and positioning solution for B2B/B2C segments

- Developed comprehensive business plan including SWOT analysis, competitive positioning against established players, pricing strategy ($50-80 per unit), and go-to-market approach targeting smart cities, schools, and health-conscious consumers
- Applied data-driven decision-making frameworks using real-time AQI analytics to demonstrate business value proposition, securing stakeholder buy-in and presenting entrepreneurial case for product commercialization with 2-year break-even projection

Cybercrime and Engineer’s Ethical Role CASE STUDY OF PROFESSIONAL VALUES AND ETHICS

Cybercrime and Engineer's Ethical Role – Professional Values & Ethics Case Study

- Analyzed real-world cybercrime cases (data breaches, ransomware attacks, social engineering) examining engineers' ethical responsibilities under IEEE/ACM Code of Ethics and professional accountability in preventing security vulnerabilities
- Evaluated ethical dilemmas involving whistleblowing, responsible disclosure, dual-use technologies, and conflicts between organizational pressure and public safety, applying ethical decision-making frameworks (utilitarian, deontological, virtue ethics)
- Developed recommendations for ethical engineering practices including secure-by-design principles, privacy considerations, and professional obligations to report security flaws while balancing stakeholder interests and legal constraints

Digital Filter Design and Implementation OF Digital signal Processing

Digital Filter Design and Implementation – Digital Signal Processing

- Designed and analyzed FIR and IIR digital filters (lowpass, highpass, bandpass, bandstop) using MATLAB, applying window methods (Hamming, Kaiser), frequency sampling, and bilinear transformation techniques to meet specified frequency response requirements
- Implemented filter algorithms in C/Python calculating difference equations for real-time signal processing, evaluating performance metrics including cutoff frequency, passband ripple, stopband attenuation, and computational complexity (multiply-accumulate operations)
- Validated filter characteristics through frequency response analysis, pole-zero plots, and impulse response testing, demonstrating noise reduction, signal enhancement, and spectral shaping applications on audio/communication signals

Traffic Control System OF Digital System Design

Traffic Control System – Digital System Design

- Designed intelligent traffic light controller using finite state machine (FSM) architecture with sequential logic circuits, implementing state transitions for multi-directional traffic flow with pedestrian crossing and emergency vehicle priority modes
- Developed system using Verilog/VHDL on FPGA (Xilinx/Altera) or discrete logic ICs (counters, flip-flops, multiplexers), incorporating timing circuits with adjustable delay periods (15-60 seconds) and sensor inputs for vehicle detection
- Simulated and verified functionality using ModelSim/Vivado testbenches, successfully implementing 4-way intersection control with synchronization logic preventing conflicting green signals and achieving <1ms state transition response time

Implementation of Page Replacement Algorithm OF Operating Systems

Implementation of Page Replacement Algorithms – Operating Systems

- Implemented and compared classical page replacement algorithms including FIFO, LRU (Least Recently Used), Optimal, and Clock/Second Chance using C/C++/Python, simulating virtual memory management with configurable page frame sizes and reference strings
- Developed simulation framework to analyze algorithm performance metrics including page fault rate, hit ratio, and memory access patterns across varying workloads, demonstrating Belady's anomaly in FIFO and optimal performance characteristics of LRU
- Visualized algorithm behavior through detailed trace outputs and statistical analysis, achieving 25-40% reduction in page faults with LRU compared to FIFO on real-world reference string datasets, validating theoretical complexity and practical efficiency tradeoffs

Prayer Alarm OF MOBILE APPLICATION SYSTEM

Prayer Alarm – Mobile Application System

- Developed cross-platform prayer reminder application using Flutter/React Native with automated Salah time calculations based on GPS location, implementing astronomical algorithms for accurate Fajr, Dhuhr, Asr, Maghrib, and Isha timings across global time zones
- Integrated push notifications, customizable Adhan sounds, Qibla direction compass using device sensors (gyroscope, magnetometer), and Islamic calendar with Hijri date conversion providing comprehensive worship management features
- Designed user-friendly interface with personalized alarm settings, snooze functionality, mosque finder using Google Maps API, and offline mode supporting 1000+ cities, achieving 4.5+ star rating with 10K+ downloads on Play Store/App Store

TEMPERATURE AND HUMDITY SENSPR WITH ARDUINO OF MICROPROCESSOR AND INTERFACING TECHNIQUES

Temperature and Humidity Sensor with Arduino – Microprocessor and Interfacing Techniques

- Interfaced DHT11/DHT22 digital sensor with Arduino Uno using single-wire serial communication protocol, implementing sensor library for real-time temperature (±2°C accuracy) and humidity (±5% RH) measurements with 2-second sampling intervals
- Developed embedded C++ program to read sensor data via GPIO pins, process digital signals, and display readings on 16x2 LCD using I2C interface and serial monitor for debugging, demonstrating proficiency in peripheral interfacing and bus protocols
- Implemented data logging system storing sensor readings to SD card/EEPROM with timestamp functionality, creating temperature/humidity monitoring solution applicable for weather stations, greenhouse automation, and HVAC control systems

GOCERY MANAGMENT SYSTEM OF SOFTWARE ENGINEERING

Grocery Management System – Software Engineering

- Developed full-stack grocery store management application using Agile/Waterfall SDLC methodology, implementing inventory management, point-of-sale (POS), billing, supplier tracking, and sales analytics modules with MySQL database backend
- Applied software engineering principles including requirement analysis, UML modeling (use case, class, sequence diagrams), modular design patterns (MVC architecture), and rigorous testing (unit, integration, system testing) ensuring 95%+ bug-free deployment
- Created user-friendly interface using Java/Python/C# with features for stock level monitoring, automated reorder alerts, barcode scanning, customer management, and daily sales reporting, reducing inventory discrepancies by 30% and checkout time by 40%

ONLINE LIBRARY MANAGMENT SYSTEM OF DATABASES SYSTEM

Online Library Management System – Database Systems

- Designed and implemented comprehensive library database using MySQL/PostgreSQL with normalized schema (3NF/BCNF), creating ER diagrams and relational models for entities including books, members, transactions, authors, and categories with proper primary/foreign key relationships
- Developed complex SQL queries implementing CRUD operations, multi-table joins, aggregate functions, subqueries, and stored procedures for functionalities like book search, issue/return tracking, fine calculation, overdue management, and inventory reporting
- Built web-based interface using PHP/Python/Java connecting to database via JDBC/PDO, implementing transaction management with ACID properties, indexing for query optimization, and user authentication achieving <200ms query response time for 10K+ book records

Financial Institution System Network Design OF COMPUTER COMMUNICATION AND NETWORKS

Financial Institution System Network Design – Computer Communication and Networks

- Designed scalable three-tier hierarchical network architecture for banking institution using Cisco Packet Tracer/GNS3, implementing core-distribution-access layer topology with redundant links, VLAN segmentation for departments (Accounts, Loans, IT, Customer Service), and inter-VLAN routing for secure data flow

- Configured advanced networking protocols including OSPF/EIGRP for dynamic routing, STP/RSTP for loop prevention, HSRP/VRRP for gateway redundancy, and implemented security measures with ACLs, port security, DHCP snooping, and firewall rules protecting against unauthorized access
- Deployed enterprise network supporting 500+ users with QoS for prioritizing critical transactions, DMZ for web servers, VPN (IPSec/SSL) for remote branch connectivity, and achieved 99.9% network uptime through dual ISP links and backup power considerations

GALAXY TAB S8 OF COMPUTER AIDED DESIGN

Galaxy Tab S8 CAD Modeling – Computer Aided Design

- Created detailed 3D model of Samsung Galaxy Tab S8 tablet using SolidWorks/AutoCAD/CATIA, applying advanced modeling techniques including surface modeling, filleting, chamfering, and parametric design to replicate device dimensions (253.8 × 165.3 × 6.3mm) with high accuracy
- Developed complete assembly with 25+ individual components (display panel, housing, buttons, camera module, S-Pen stylus, speakers) using constraints, mates, and boolean operations, generating exploded views and bill of materials (BOM) for manufacturing visualization
- Produced professional engineering drawings with orthographic projections, sectional views, dimensioning, tolerances (GD&T), and photorealistic renderings using KeyShot/V-Ray demonstrating proficiency in CAD standards and technical documentation

Speech Recognition System OF SIGNAL AND SYSTEMS

Speech Recognition System – Signals and Systems

- Developed speech recognition system using MATLAB/Python applying signal processing techniques including pre-emphasis filtering, frame blocking, windowing (Hamming/Hanning), and FFT-based spectral analysis to extract acoustic features from audio signals sampled at 16kHz
- Implemented Mel-Frequency Cepstral Coefficients (MFCC) feature extraction with 13 coefficients, delta/delta-delta features, and trained classification models using Hidden Markov Models (HMM)/Neural Networks achieving 85-95% recognition accuracy on isolated word dataset
- Applied systems theory concepts including convolution, Z-transform for filter design, LTI system analysis, and frequency domain processing to denoise speech signals, enhance voice quality, and distinguish between phonemes for pattern recognition

4-BIT MICROPROCEEESOR OF COMPUTER ARCHITECTURE AND ORGANIZATION

4-Bit Microprocessor – Computer Architecture and Organization

- Designed complete 4-bit microprocessor architecture implementing Von Neumann/Harvard model with components including 4-bit ALU (ADD, SUB, AND, OR, XOR operations), accumulator, program counter, instruction register, control unit using finite state machine, and 16-word memory
- Developed custom instruction set architecture (ISA) with 8-12 instructions supporting arithmetic, logical, data transfer (LOAD, STORE), and control flow operations (JUMP, BRANCH), implementing fetch-decode-execute cycle with multi-cycle datapath timing
- Implemented and simulated design using Logisim/Verilog/VHDL demonstrating functional CPU executing assembly programs, achieving 5-10 clock cycles per instruction with successful validation through test programs (addition, factorial, Fibonacci sequence calculations)

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4-Bit Microprocessor – Computer Architecture and Organization

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CRICKET SCORE SHEET SYSTEM OF DATA STRUCTURES AND ALGORTHIMS

Cricket Score Sheet System – Data Structures and Algorithms

- Developed comprehensive cricket scoring application using C++/Java implementing linked lists for ball-by-ball commentary tracking, hash tables for O(1) player lookup, binary search trees for sorted leaderboards, and dynamic arrays for managing team rosters of 11+ players
- Designed algorithms for real-time score calculation, run rate computation, bowling/batting statistics (average, strike rate, economy rate), and implemented efficient searching (binary search for player records) and sorting (quicksort/mergesort for rankings by performance metrics)
- Created menu-driven system with functionalities including innings management, wicket tracking, extra runs handling (wides, no-balls, byes), partnership analysis, and match summary generation achieving O(log n) time complexity for critical operations on datasets containing 1000+ match records

HANGMAN GAME C++ OF OBJECT ORIENTED PROGRAMMING

Hangman Game in C++ – Object Oriented Programming

- Developed interactive word-guessing game applying core OOP principles including encapsulation (private data members for word, guessed letters, lives), inheritance (base Game class with derived DifficultyLevels), polymorphism (virtual functions for hint systems), and abstraction through well-defined class interfaces
- Implemented game mechanics using C++ STL containers (vectors for word lists, sets for tracking guessed characters), file I/O for loading dictionary files with 500+ words, string manipulation algorithms, and random number generation for word selection across multiple difficulty levels
- Designed modular architecture with classes for Player, GameEngine, WordDatabase, and Display, utilizing constructors/destructors for resource management, operator overloading for score comparison, and exception handling for invalid user inputs achieving robust gameplay experience

RAIN DETECTION SYSTEM OF DIGITAL LOGIC SYSTEM

Rain Detection System – Digital Logic Design

- Designed automated rain detection circuit using combinational logic (AND, OR, NOT gates) and sequential logic (D flip-flops, counters) to process rain sensor input, implementing threshold detection with comparators and triggering output devices (buzzer alarm, relay for window closure, LED indicators)
- Developed Boolean expressions, truth tables, and K-maps for logic minimization achieving optimal gate-level implementation with 10-15 logic gates, simulated circuit in Proteus/Logisim validating functionality with various rainfall intensity scenarios
- Implemented complete system with rain sensor interfacing, debouncing circuits for noise immunity, timer circuits using 555 IC for delayed response (preventing false triggers), and 7-segment display showing detection status for home/vehicle automation applications

FM LONG RANGE TRANSMISSION OF ENGINEERING WORKSHOP

FM Long Range Transmission – Engineering Workshop

- Designed and constructed FM radio transmitter circuit operating at 88-108 MHz frequency band using LC oscillator (Colpitts/Hartley), varactor diode modulator, RF power amplifier (2N2222/2N3866 transistor), and quarter-wave antenna achieving 1-3 km transmission range with clear audio quality
- Demonstrated practical engineering skills including PCB soldering, component selection (capacitors, inductors, resistors), impedance matching for antenna coupling, voltage regulation (9V battery with 7805 regulator), and troubleshooting using oscilloscope and spectrum analyzer
- Applied frequency modulation theory calculating carrier frequency, modulation index, bandwidth (Carson's rule), and tested transmission performance across various distances validating wireless communication principles while adhering to low-power broadcasting regulations

Traffic Control System OF ELECTRONIC DEVICES AND CIRCUITS

Traffic Control System – Electronic Devices and Circuits

- Designed and assembled traffic light controller circuit using 555 timer ICs in astable/monostable mode for precise timing intervals (15-60 seconds), NE555/CD4017 decade counter for sequential LED switching, and BJT/MOSFET transistor switches driving high-current traffic signal loads
- Implemented multi-phase traffic signal timing using discrete electronic components including resistors (current limiting for LEDs), capacitors (RC timing networks), diodes (flyback protection), and voltage regulators (LM7805/7812) ensuring stable 5V/12V power supply operation
- Constructed and tested complete system on breadboard/PCB demonstrating transistor switching characteristics, LED current calculations (20mA typical), relay driver circuits for AC lamp control, and emergency vehicle override functionality using priority interrupt circuitry

LU DECOMPOSION FILE HANDLING OF PROGRAMMING HANDLING

LU Decomposition with File Handling – Programming Fundamentals

- Implemented LU decomposition algorithm in C++/Python/Java to factorize square matrices into lower triangular (L) and upper triangular (U) matrices using Doolittle's/Crout's method, solving systems of linear equations ($Ax = b$) through forward and backward substitution with $O(n^3)$ complexity

- Developed comprehensive file handling module using fstream/file I/O operations to read coefficient matrices and constant vectors from text/CSV files, perform LU factorization, and write decomposed matrices, determinant calculations, and solution vectors to output files with formatted display
- Applied programming concepts including 2D array/vector manipulation, nested loops for matrix operations, pivoting strategies for numerical stability, exception handling for singular matrices, and modular function design achieving accurate solutions for 10x10+ matrix systems

Languages

Urdu, English